GENERATIVE & AGENTIC AI SYSTEMS · CAPSTONE

BankWise AI

A 24/7 banking assistant for Egyptian customers, in Arabic

Problem statement and proposed solution

Problem statement

Everyday banking questions still send customers to a branch or onto a call-center line.

Branch queues

Customers wait in person for simple answers about fees, products and loans.

Call-center waits

Hold times are inconsistent, and support depends on working hours.

Repeated questions

Support teams answer the same eligibility and fee questions again and again.

Proposed solution

BankWise AI is a virtual assistant that answers in Egyptian Arabic at any hour.

Loan eligibility

A first-pass check with a clear explanation of the result.

FAQ and fees

Answers grounded in bank policy documents and fee schedules.

Safe by design

Estimates only, never final approval. Rate limiting and clear fallback messages.

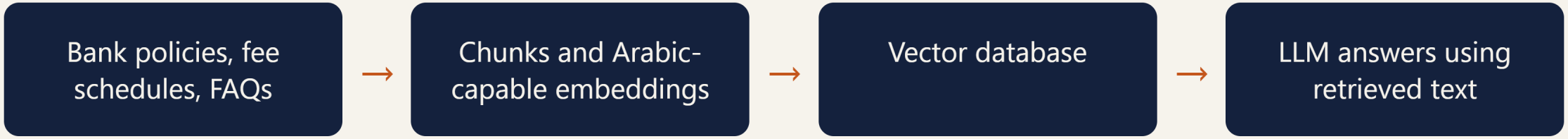
Feature 1: Loan eligibility

1. Customer gives income, debts, loan amount, tenor and age
2. A rule-based tool applies the rules: debt burden ratio, minimum income, maximum tenor, age limits
3. The tool returns eligible or not, the reasons, and the estimated installment
4. The assistant explains the result in plain language

Why rule-based

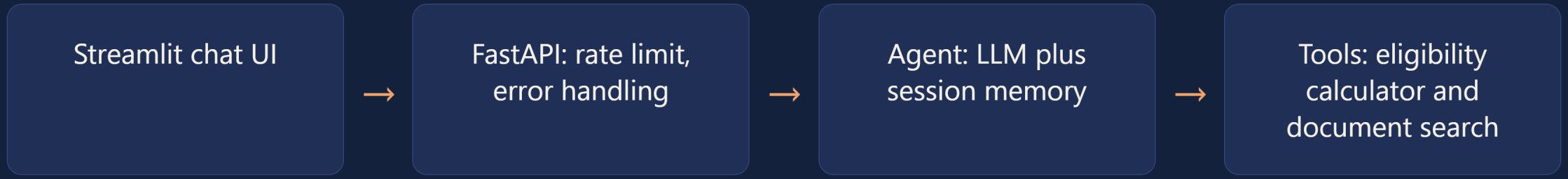
Exact math instead of LLM guesswork. Every rule traces to a published source and lives in one config file.

Feature 2: FAQ with RAG



Egyptian Arabic questions are matched against Arabic bank documents, and each answer stays tied to its source.

System architecture



A monitoring dashboard tracks top questions, latency, tokens and cost per conversation. BankWise AI is designed to embed in the bank's own app, where the bank's system handles customer login.

Scope, metrics and next steps

In scope now

Eligibility tool, FAQ with RAG, chat UI, rate limiting, error handling, monitoring dashboard.

Success metrics

8 of 10 sample questions answered correctly. Installments match independent calculations. Zero unhandled exceptions on scripted tests.

Future work

Bilingual support (Arabic and English). Complaint routing with category classification. Integration with the bank's authenticated app for account-specific questions. Optional ML eligibility model.